



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FLASH COLUMN OVERHEAD VAPOR CHILLER (E-2310)

Docu. No. : 14780-8120-PS-E2310
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
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TITLE :

FLASH COLUMN OVERHEAD VAPOR CHILLER

(E-2310)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-E2310
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 11, 2016	AS BUILT		S.W.Park	J.S.Chung	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE		S.W.Park	J.S.Chung	S.W.Lee
REV. NO.	DATE	DESCRIPTION		PREP'N	REVIEW	QA REVIEW
REV. NO.	PREPARATION DEPT.	PREP'N	REVIEW	QA REVIEW		APPROVAL
Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
		MAY 11, 2016	MAY 11, 2016	-		MAY 11, 2016

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SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

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Case		Design	TEMA Type	AEU	Orientation	Horizontal	Rev
Number of Units		1	Shells per Unit	1	Shells in Series	1	
Surface per Unit	m ²	224.4	Surface per Shell	m ²	Shells in Parallel	1	
PERFORMANCE OF ONE UNIT							
		In	SHELLSIDE	Out	In	TUBESIDE	Out
Fluid Name		Hydrocarbon			Chilled Water (Note 4)		
Total Flowrate (Note 2)		kg/h	3085 × 1.1		64472 × 1.1		
Vapour		kg/h	3027 × 1.1	231.4 × 1.1	0		0
Liquid		kg/h	0	2795.6 × 1.1	64472 × 1.1		64472 × 1.1
Fluid Condensed		kg/h	2795.6 × 1.1		0		
Fluid Evaporated		kg/h	0		0		
Noncondensibles			58 × 1.1				
LIQUID PROPERTIES							
		SEE PAGES 7, 8			SEE PAGE 9		
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES							
		SEE PAGES 7, 8					
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	38.0	5.0		0.1	5.0	
Operating Pressure	kg/cm ² g	-1.0163			6.75		
Number of Passes per Shell		1			8		
Average Velocity	m/s	16.8			1.0		
Pressure Drop (Allow / Calc)	kg/cm ²	0.00141 (Note 1)	0.00169 (Note 1)		0.703 (Note 20)	0.714	
Fouling Factor	h.m ² .°C/kcal	0.0003			0.0004		
Duty (Note 2)	kcal/h × 10 ⁶	0.2927 × 1.1			MTD	°C	18.6
U-value Service/Clean	kcal/h.m ² .°C	82	88		Over Surface	%	6.5
DESIGN CONDITIONS							
		SHELLSIDE			TUBESIDE		
Design pressure max/min	kg/cm ² g	4.5			11.5		
Full Vacuum		Required	at 38 °C		Not Required		
Test Pressure	kg/cm ² g	To Code			To Code		
Design Temperature max/min	°C	120	-9		120	-14	
Corrosion Allowance	mm	1.5			3	(for KCS)	
Insulation Details		Cold Conservation			Cold Conservation		
Hydrogen Service		No			No		
Sour Service		No			No		
Lethal Service		No			No		
Cyclic Service		No			No		
Cleaning Method		Mechanical			Mechanical		
Stress Relief for Process Reasons		No			No		
Heat Release Curve		Yes			Yes		

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CONSTRUCTION OF ONE SHELL (Note 3)

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SHELL	Shell I.D.	mm	1280	Channel I.D. (If different)	mm	
TUBES	Number of Tubes		780 (Note 15)	Tube Length	mm	4877
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	28.575	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded (Note 11)	Tube-Baffle Diam. Clearance	mm	0.4
BAFFLES	Number of Baffles		6	Baffle Type		Double Seg.
	Baffle Cut	%	27.68	Baffle Spacing (Note 5)	mm	600 (c/c)
	Baffle Cut Orientation		Vertical	Double Seg. Baffle Overlap	mm	57.15
	Baffle Thick	mm	12.7	Baffle-Shell Diam. Clearance	mm	6.4
BUNDLE	Impingement Protection		Rods (Note 9)	Tube Supports (Note 6)		U-bend Tan Line
	Pairs of Sealing Strips		1	Sealing Strip Details (Note 7, 21)		First to Last Baffle
	Passlane Sealing		Yes	Passlane Sealing Details (Note 7)		First to Last Baffle
	U-Bend Diam.	mm	57.15	U-Bend Orientation		Horizontal
	U-Bend Area Included		No	U-Bend Support Details		(Note 13)
	Longitudinal baffle		No	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE

Service	No.	N.B.		Rating	Facing
Inlet S1	1	30	Inch	150	RFWN
Outlet S2	1	18	Inch	150	RFWN
Outlet S3	1	3	Inch	150	RFWN
Vent SV	1	2	Inch	150	RFLWN
Drain SD	1	2	Inch	150	RFLWN

TUBESIDE

Service	No.	N.B.		Rating	Facing
Inlet T1	1	6	Inch	150	RFWN
Outlet T2	1	6	Inch	150	RFWN
Vent TV	1	2	Inch	150	RFLWN
Drain TD	1	2	Inch	150	RFLWN

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	304L Stainless Steel (UOP Std 3-15)
Floating Head Cover	N.A.
Channel / Channel Cover	Killed Carbon Steel (UOP Std 3-11)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	N.A.
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: 304L Stainless Steel
Nozzles / Nozzle Flange	Tubeside: Killed Carbon Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell (Tonnes)	Exchanger Weight, Dry	11.3	Exchanger Weight, Wet	19.6
	Bundle Weight, Dry	4.4		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-12-4, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

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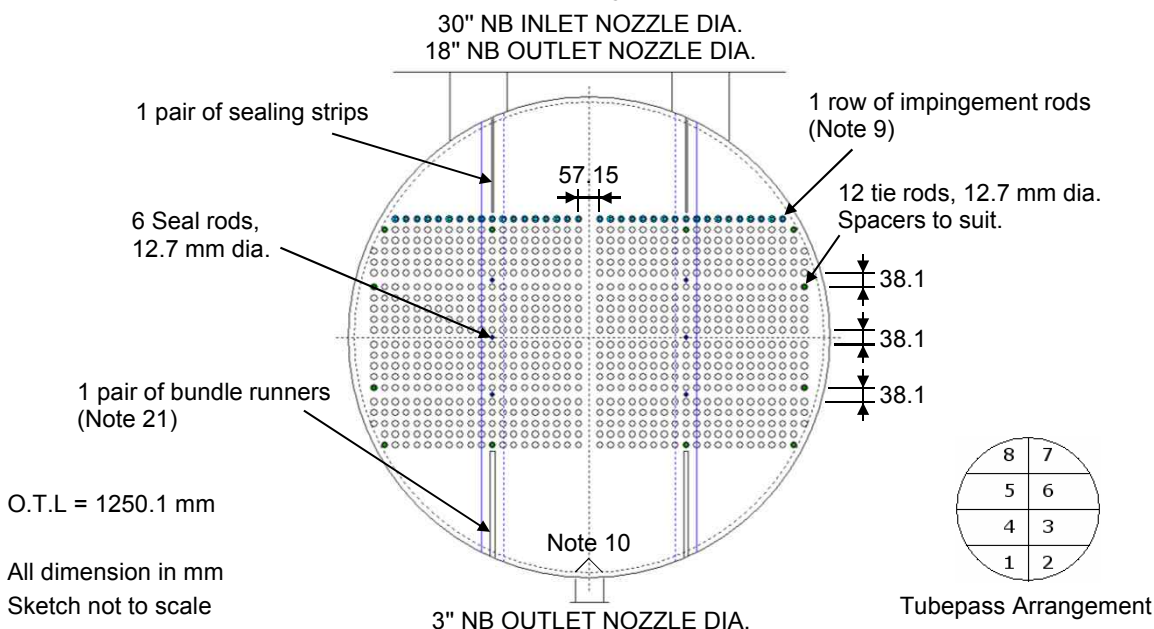
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Notes:

1. The allowable shellside pressure drop to be confirmed by CONTRACTOR. At normal flow, excluding 10% design margin, the calculated shellside pressure drop is 0.00141 kg/cm².
2. Flow and duty include 10% design margin.
3. Item to be provided by Contration Vacuum Producing Equipment VENDOR X-2301. Refer to UOP 963766-511.
4. Chilled Water properties are based on 20 wt% Ethylene Glycol and are to be confirmed by the CONTRACTOR.
5. The inlet baffle spacing is 1247.5 mm.
6. A full support plate at U-bend shall be provided.
7. Sealing strips and seal rods shall not restrict the bundle cleaning lanes.
8. Hydrocarbon inlet line size of 30 inch to be confirmed by CONTRACTOR. Equipment vendor to confirm the shellside inlet nozzle size if increases.
9. Provide one row of impingement rods with the same diameter, same pitch, and same layout as the tubes.
10. All single baffles extending to the bottom of the shell shall be provided with V-notch at the lowest point.
11. Strength-weld tubes to tubesheet per requirements on page 10.
12. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
13. Provide one inbend U-bend support to hold 8 outer most tube rows.
14. Vapor and liquid will exit shell via separate nozzles. Vapor line size is 18 inch and liquid line size is 2 inch. Item shall be designed to allow for vapor-liquid separation.
15. For U-tube design, Number of tubes equals number of straights.
16. Elevate the exchanger for free draining on shell to the Decanter in the Oxidation Section. CONTRACTOR to advise and confirm.
17. Provide 2 intermediate support plates at the inlet baffle spacing.
The maximum unsupported tube span of 1017 mm.
The intermediate support plates shall be extended to hold the impingement rods.
18. The 1st# single baffle shall be designed for attachment of all impingement rod.
19. Test ring shall be provided by VENDOR when required.
20. The allowable tubeside pressure drop to be confirmed later.
21. Bundle runners and sealing strips shall not restrict the shellside entrance and exit flow paths.
22. Deleted.

Tube Layout



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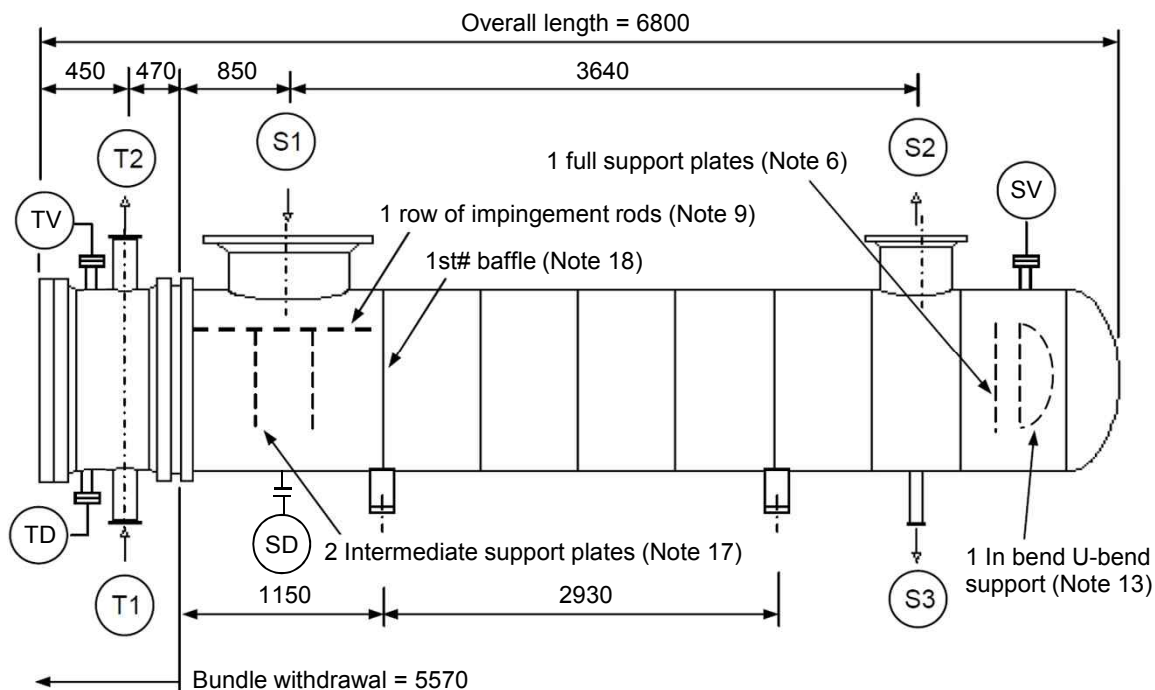
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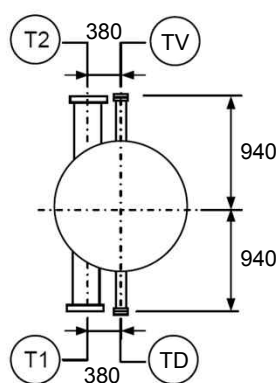
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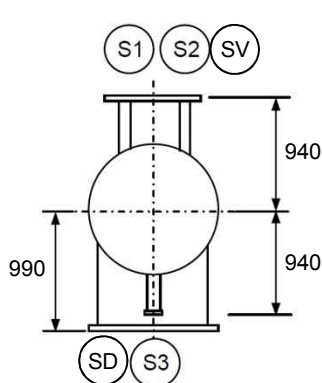
Setting Plan



Tubeside Nozzle



Shellside Nozzle



Shell ID : 1280 mm

Tube length : 4877 mm

All dimension in mm
Sketch not to scale

PROCESS NOZZLE SUMMARY

REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	Hydrocarbon Inlet	30"	150#RFLWN		
S2	Hydrocarbon Vapor Outlet	18"	150#RFLWN		
S3	Hydrocarbon Liquid Outlet	3"	150#RFLWN		
T1	Chilled Water Inlet	6"	150#RFLWN		
T2	Chilled Water Outlet	6"	150#RFLWN		
SV	Shellside Vent	2"	150#RFLWN		CW/blind
SD	Shellside Drain	2"	150#RFLWN		CW/blind
TV	Tubeside Vent	2"	150#RFLWN		CW/blind
TD	Tubeside Drain	2"	150#RFLWN		CW/blind

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HotSide Property Curve 1

Pressure -1.0140 kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
52.9	104.28	1.0000	0.0609	0.007	0.0093	0.32
45.6	101.92	1.0000	0.0625	0.007	0.0087	0.32
38.1	99.56	1.0000	0.0641	0.007	0.0080	0.31
37.4	85.34	0.8408	0.0625	0.007	0.0081	0.31
36.5	71.12	0.6828	0.0593	0.007	0.0082	0.31
35.1	56.89	0.5270	0.0561	0.008	0.0084	0.31
32.8	42.67	0.3756	0.0513	0.008	0.0087	0.31
28.5	28.45	0.2338	0.0432	0.009	0.0094	0.31
18.8	14.22	0.1164	0.0320	0.010	0.0110	0.31
-3.1	0.00	0.0527	0.0192	0.012	0.0137	0.32

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
52.9	-	-	-	-	-	-	-
45.6	-	-	-	-	-	-	-
38.1	847.2	0.626	0.104	0.42	26.3	32.7	358
37.4	847.6	0.630	0.104	0.42	26.4	32.7	358
36.5	848.4	0.637	0.104	0.41	26.5	32.7	358
35.1	849.5	0.648	0.104	0.41	26.6	32.7	358
32.8	851.4	0.667	0.105	0.41	26.9	32.8	358
28.5	855.0	0.705	0.105	0.41	27.3	32.8	358
18.8	863.0	0.804	0.107	0.39	28.4	32.8	358
-3.1	881.2	1.134	0.111	0.37	30.9	33.3	358

HotSide Property Curve 2

Pressure -1.0158 kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
52.9	104.28	1.0000	0.0545	0.007	0.0093	0.32
44.6	101.60	1.0000	0.0561	0.007	0.0086	0.32
36.0	98.92	1.0000	0.0561	0.007	0.0079	0.31
35.4	84.79	0.8421	0.0561	0.007	0.0079	0.31
34.5	70.66	0.6854	0.0529	0.007	0.0081	0.31
33.1	56.53	0.5309	0.0513	0.008	0.0082	0.31
30.9	42.40	0.3804	0.0464	0.008	0.0086	0.31
26.8	28.26	0.2389	0.0400	0.009	0.0092	0.31
17.6	14.13	0.1204	0.0288	0.010	0.0108	0.31
-3.5	0.00	0.0540	0.0176	0.012	0.0136	0.32

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
52.9	-	-	-	-	-	-	-
44.6	-	-	-	-	-	-	-
36.0	848.9	0.642	0.104	0.41	26.6	32.7	358
35.4	849.3	0.646	0.104	0.41	26.6	32.7	358
34.5	850.1	0.653	0.104	0.41	26.7	32.7	358
33.1	851.2	0.664	0.105	0.41	26.8	32.7	358
30.9	853.0	0.683	0.105	0.41	27.1	32.8	358
26.8	856.4	0.721	0.106	0.40	27.5	32.8	358
17.6	864.0	0.817	0.107	0.39	28.5	32.8	358
-3.5	881.4	1.141	0.111	0.37	30.9	33.2	358

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HotSide Property Curve 3

Pressure -1.0175 kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
52.9	104.28	1.0000	0.0481	0.007	0.0093	0.32
43.5	101.27	1.0000	0.0497	0.007	0.0085	0.32
33.8	98.26	1.0000	0.0513	0.007	0.0077	0.31
33.2	84.22	0.8435	0.0497	0.007	0.0078	0.31
32.3	70.18	0.6882	0.0481	0.007	0.0079	0.31
31.0	56.15	0.5349	0.0448	0.008	0.0081	0.31
28.9	42.11	0.3854	0.0416	0.008	0.0084	0.31
25.0	28.07	0.2444	0.0352	0.009	0.0090	0.30
16.4	14.04	0.1248	0.0256	0.010	0.0105	0.31
-3.9	0.00	0.0554	0.0160	0.012	0.0134	0.32

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
52.9	-	-	-	-	-	-	-
43.5	-	-	-	-	-	-	-
33.8	850.7	0.660	0.104	0.41	26.8	32.7	358
33.2	851.1	0.664	0.105	0.41	26.8	32.7	358
32.3	851.8	0.671	0.105	0.41	26.9	32.7	358
31.0	852.9	0.682	0.105	0.41	27.0	32.7	358
28.9	854.6	0.701	0.105	0.41	27.3	32.8	358
25.0	857.9	0.738	0.106	0.40	27.7	32.8	358
16.4	865.0	0.833	0.108	0.39	28.7	32.8	358
-3.9	881.7	1.149	0.111	0.37	31.0	33.2	358

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ColdSide Property Curve

Pressure

6.750

kg/cm²g

Temperature	Entire Stream	Entire Stream	Vapour	Vapour	Vapour	Vapour
	Enthalpy	Mass Vapour	Density	Viscosity	Therm. Cond.	Spec. Heat
°C	kcal/kg	fraction	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C
0.1	0.00	0.0000	-	-	-	-
0.6	0.50	0.0000	-	-	-	-
1.2	1.01	0.0000	-	-	-	-
1.7	1.51	0.0000	-	-	-	-
2.3	2.02	0.0000	-	-	-	-
2.8	2.52	0.0000	-	-	-	-
3.4	3.02	0.0000	-	-	-	-
3.9	3.53	0.0000	-	-	-	-
4.5	4.03	0.0000	-	-	-	-
5.0	4.54	0.0000	-	-	-	-

Temperature	Liquid	Liquid	Liquid	Liquid	Liquid	Critical	Critical
	Density	Viscosity	Therm. Cond.	Spec. Heat	Surf. Tension	Pressure	Temperature
°C	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C	dyne/cm	kg/cm ² [a]	°C
0.1	1028.0	2.988	0.410	0.92	110.0	225.0	379
0.6	1027.0	2.940	0.411	0.93	110.0	225.0	379
1.2	1027.0	2.893	0.412	0.93	110.0	225.0	379
1.7	1026.0	2.847	0.413	0.93	110.0	225.0	379
2.3	1026.0	2.802	0.414	0.93	109.0	225.0	379
2.8	1025.0	2.758	0.415	0.93	109.0	225.0	379
3.4	1025.0	2.715	0.416	0.93	109.0	225.0	379
3.9	1024.0	2.672	0.417	0.93	109.0	225.0	379
4.5	1024.0	2.631	0.418	0.93	109.0	225.0	379
5.0	1023.0	2.590	0.419	0.93	108.0	225.0	379

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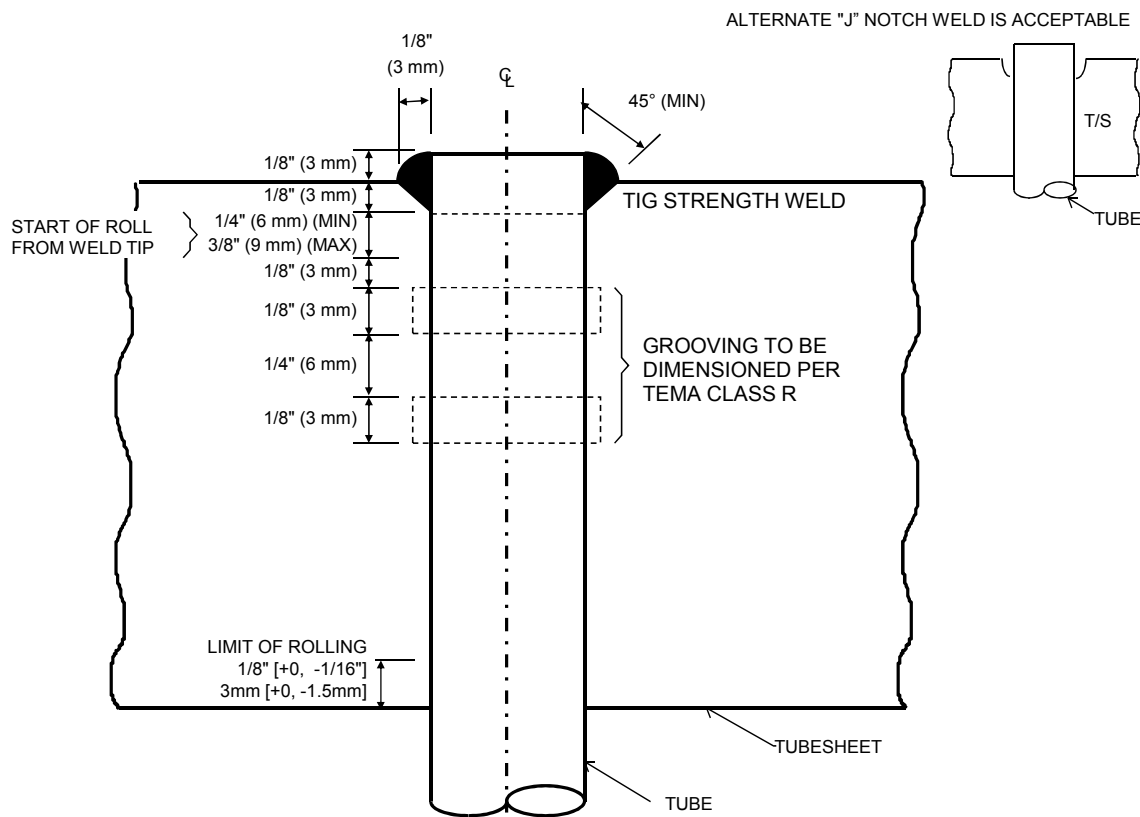
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Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

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